



Bay Ridge Residential Market

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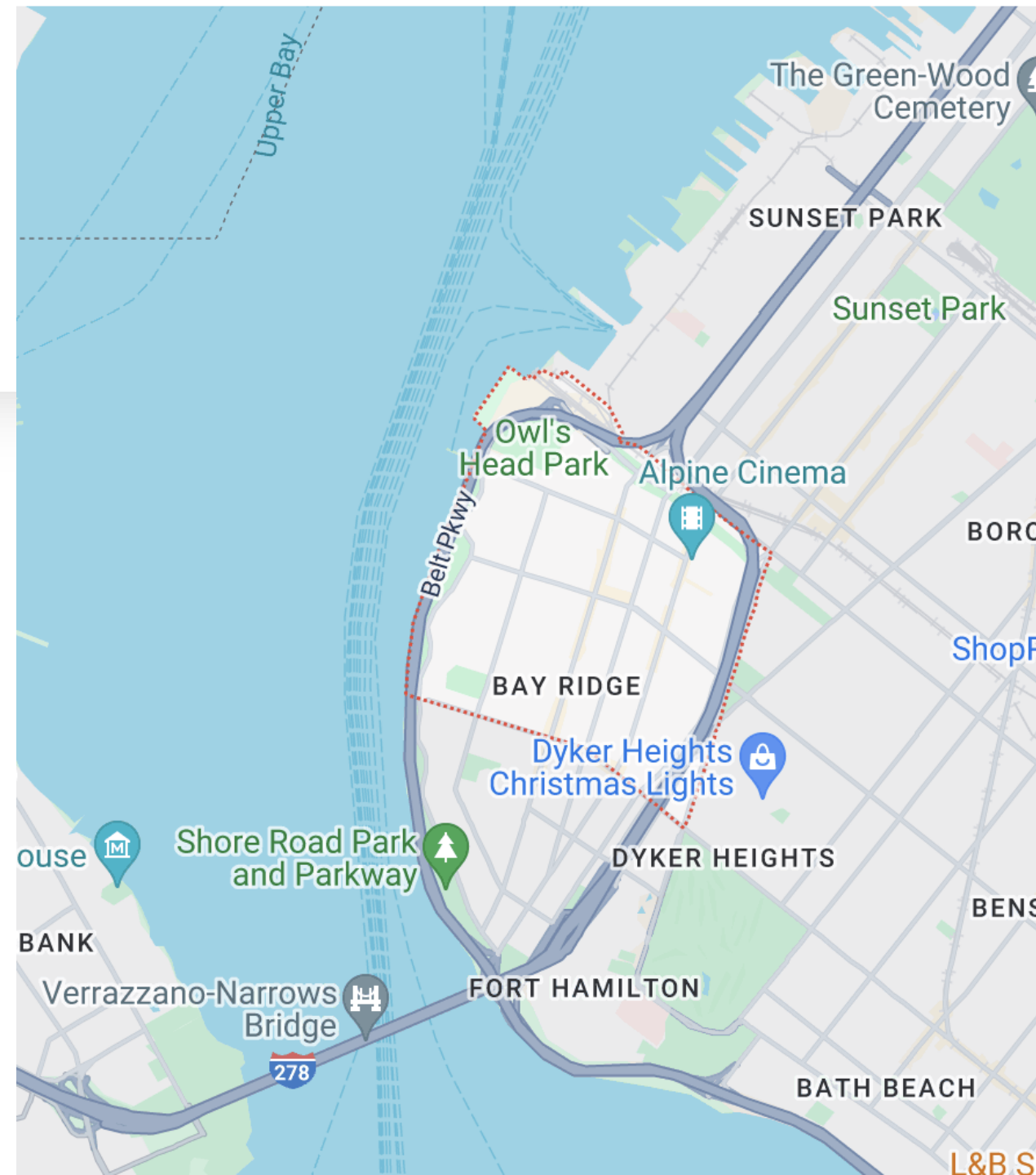
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Scenario Overview

A real estate brokerage firm is opening in the Bay Ridge, Brooklyn market. As the Business Analytics professional for our real estate brokerage firm, I've conducted an extensive analysis of residential properties and sales in Bay Ridge. **The goal is to determine if opening a new real estate office in this neighborhood is an ideal and profitable investment.**



The Best of Bay Ridge

- Offers a **variety of properties**, such as single-family homes, townhouses, condos, and multi-family buildings.
- **Affordable** property prices make it attractive for families and investors.
- **High demand for rentals** due to the neighborhood's quiet, spacious homes and access to public transportation.
- **Parks and Waterfront**, especially near Shore Road.
- **Community Feel**: multi-generational community with great schools and amenities.



Data Preparation



Use R Programming,
Excel



Correct data types
(convert text to
numbers, dates).



Connect tables.



Aggregate or
summarize data for
analysis (grouping,
Ungrouping,
pivoting).

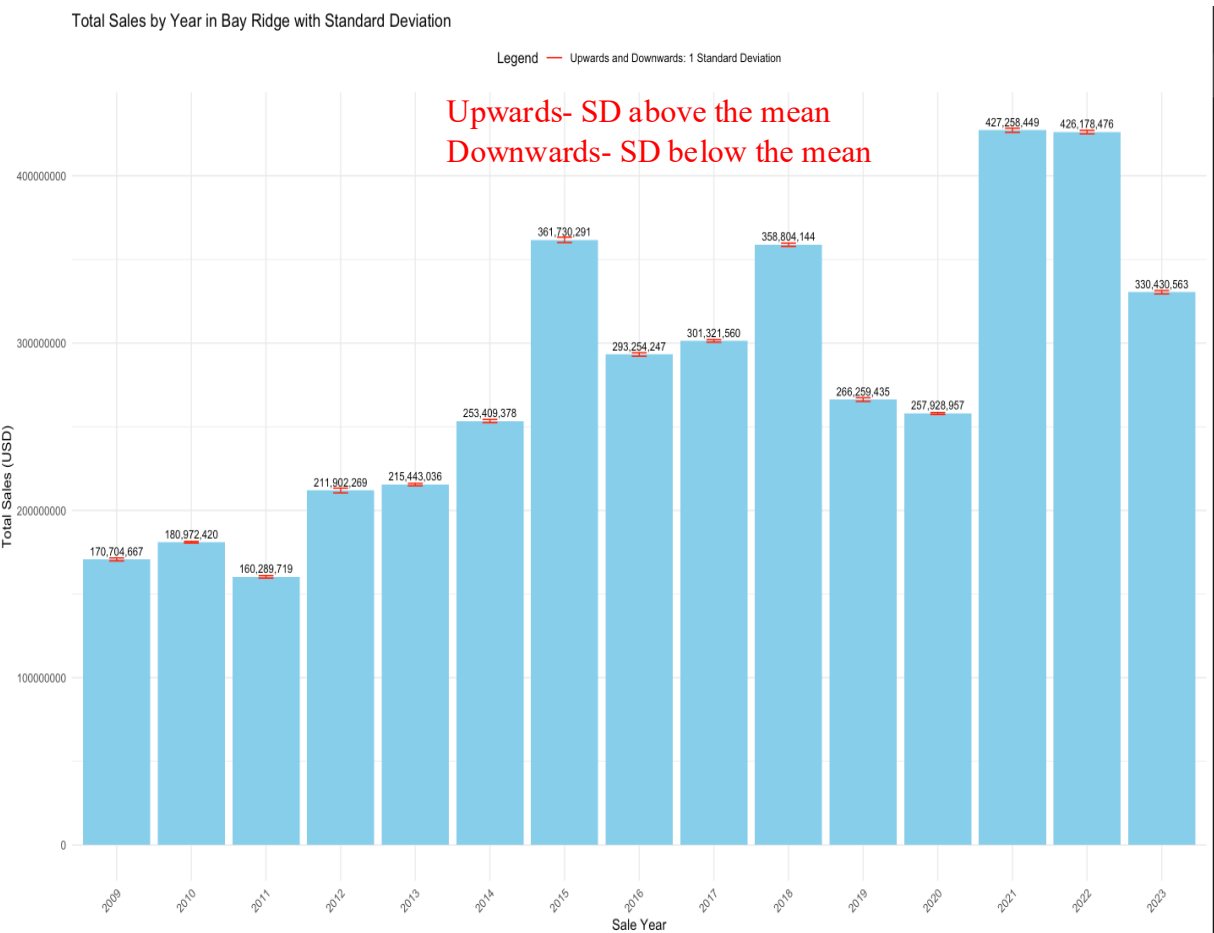


Visualize findings
using ggplot2 graphs
to identify trends
and outliers.

Descriptive Analytics

What Happened?

- The total sales increased significantly from 2009 to 2003, making \$4 billion.
- Sales in 2009 started at approximately \$170 million and peaked in 2021 at \$427 million.
- Since 2009, the standard deviation (sd) has fluctuated below and above the mean. In 2009, it was \$879,875; by 2023, it had risen to \$1,009,153. The mean sales price peak was in 2021 (\$1,424,195), but 2022 and 2023 maintained similarly high average prices.
- The mean sale price has steadily risen from \$820,696 in 2009 to \$1,436,655 in 2023, reflecting increased property value or more significant transactions.
- Since the standard deviation has fluctuated, investors can benefit from price flexibility. However, high price variability can also signal an unstable market, making it harder for investors to predict future property value.



SALE YEAR	Total Sales	MEAN SALE	SD SALE PRICE
2009	\$170,704,667.00	820,696	879,875
2010	\$180,972,420.00	773,386	484,236
2011	\$160,289,719.00	830,517	743,245
2012	\$211,902,269.00	925,337	1,422,218
2013	\$215,443,036.00	872,239	714,205
2014	\$253,409,378.00	974,651	946,934
2015	\$361,730,291.00	1,247,346	1,591,876
2016	\$293,254,247.00	1,187,264	1,051,465
2017	\$301,321,560.00	1,287,699	763,211
2018	\$358,804,144.00	1,188,093	993,278
2019	\$266,259,435.00	1,255,941	1,100,281
2020	\$257,928,957.00	1,234,110	560,461
2021	\$427,258,449.00	1,424,195	1,227,936
2022	\$426,178,476.00	1,434,944	1,054,625
2023	\$330,430,563.00	1,436,655	1,009,153

Descriptive Analytics:

Analyzing the relationship between Gross Square Foot and Sale Price

- The larger the property (more square feet), the higher the sale price.
- The red line in the graph shows the general trend: as square feet increase, the price goes up.
- The blue dots are actual sales. Most properties are smaller, but a few huge ones are sold for very high prices.
- While increasing property values tied to larger properties can bring investment and growth to the neighborhood, it also carries the risks of making the area less affordable and potentially changing the community's nature.



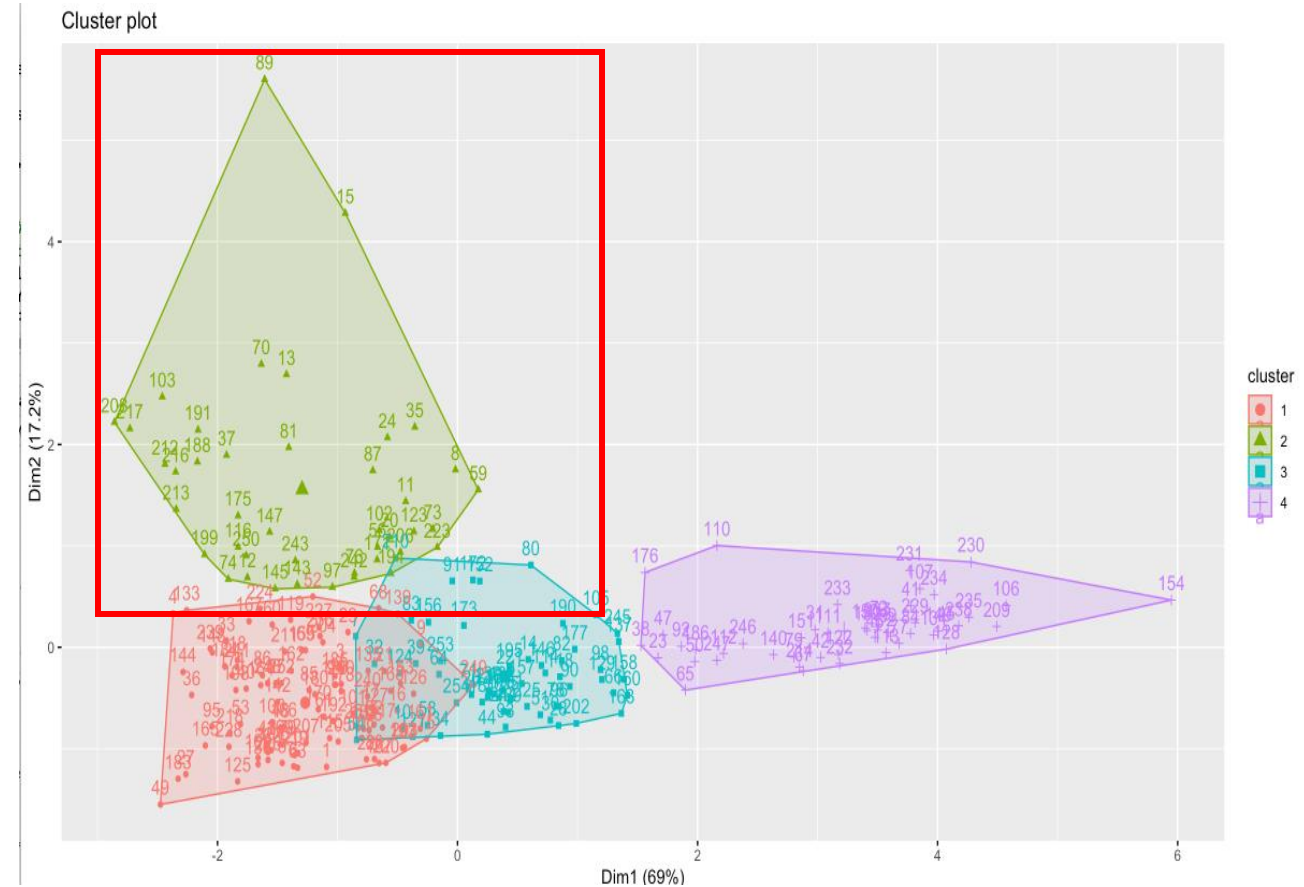
Graph Table of Content	
Sale Price (USD)	Gross Square Feet
1,200,000	2,000
1,100,000	1,800
1,000,000	1,600
950,000	1,500
850,000	1,400
800,000	1,300
750,000	1,200
700,000	1,100
650,000	1,000
600,000	900

Descriptive Analytics: K-Mean Clustering

- Bay Ridge was found in cluster two, highlighted in red.
- The median sale price for residential properties since 2009 indicates that cluster 4 has the highest median sale price, while cluster 3 has the lowest.
- Since 2009, cluster 2 has the highest residential property sales, while cluster 1 has the lowest.

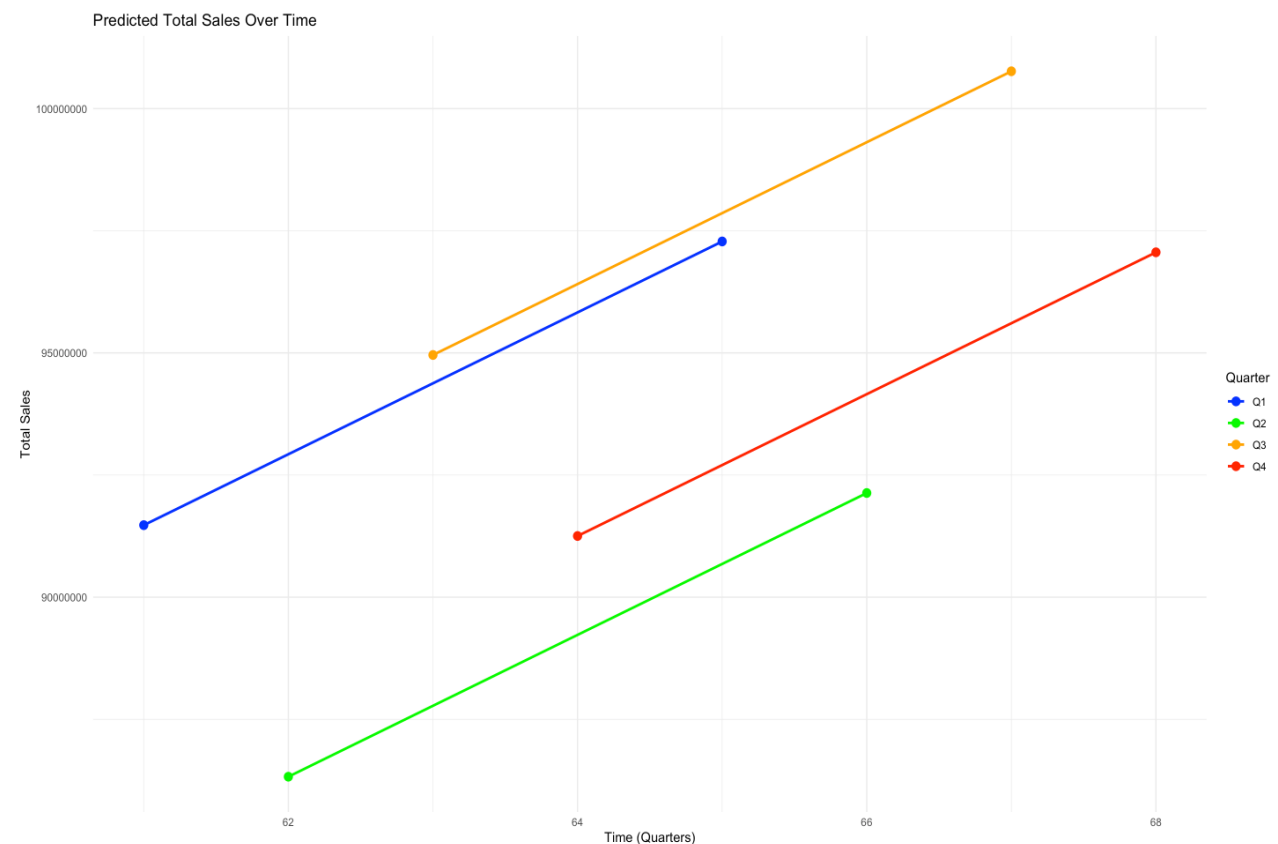
Cluster 2 insights:

- Cluster 2 has steadily growing properties but not as quickly as others. This could signal a stable investment but slower returns than faster-growing areas (like Cluster 1 or 4).
- Bay Ridge will likely fall into a moderate housing market, which could mean a consistent interest in buying properties. However, it has fewer opportunities for rapid appreciation. The real estate company should consider whether stability is worth missing out on faster-growing areas with higher potential returns.



Predictive Analytics: Multiple Regression

- Sales are projected to grow from \$91.4 million in Q1 2024 to \$100.7 million by Q3 2025.
- Predictions come with a range of uncertainty, with the lower bound starting at \$79 million in Q1 2024 and the upper bound at \$103.9 million.
- Gross square feet and sale data were the most valuable predictor variables, while the year built and residential unit was the least helpful.
- The continuous increase in predicted sales suggests that the Bay Ridge market is growing, making it a favorable time to invest in real estate.
- Since the market reflects volatility and uncertainty, real estate companies should maintain flexibility and dynamism in their strategy, balancing conservatism and growth based on market movements.



Year	Time(Quarters)	Quarters	Total Sale Prediction(fit)	Lower Bound	Upper Bond
2024	61	Q1	\$91,474,329.00	\$79,002,497.00	\$103,946,162.00
2024	62	Q2	\$86,325,263.00	\$73,853,431.00	\$98,797,095.00
2024	63	Q3	\$94,957,769.00	\$82,485,937.00	\$107,429,601.00
2024	64	Q4	\$91,252,391.00	\$78,780,559.00	\$103,724,223.00
2025	65	Q5	\$97,281,512.00	\$84,067,277.00	\$110,495,747.00
2025	66	Q6	\$92,132,446.00	\$78,918,211.00	\$105,346,681.00
2025	67	Q7	\$100,764,951.00	\$87,550,716.00	\$113,979,187.00
2025	68	Q8	\$97,059,573.00	\$83,845,338.00	\$110,273,809.00

Prescriptive Analytics: Optimization Model

- **Market Penetration:** With a commission rate of 4.50%, market penetration increases to 6.75%, boosting market share while lowering revenue per transaction.
- **Office Costs:** With an office space of 375 square feet (for one employee), monthly rent is \$3,257.44, well within the \$15,000 operating budget.
- **Staffing Costs:** Hiring one employee raises market penetration to 6.75%, balancing increased sales with manageable salary and office space costs.
- **Commission Rate Impact:** Reducing the commission to 4.50% increases market share and profitability, resulting in a projected \$1.94 million NPV over eight quarters.

Optimization Model	
Commision Rate	4.50%
Market Penetration	6.75%
Office Space	375
Office Cost	\$3,257.44
Numbers of Staffing	1 or 2
Staffing Cost	\$65,000 Annually
Montly Cost(staffing, rent, utilities)	\$8,674
NPV Revenue	\$2,132,980.42
NPV Cost	195,760.41
NPV Profit	\$1,937,220.00

Bay Ridge Life Cycle Business Model

Bay Ridge Life Cycle Business Model

Key Partners ● Real Estate Brokers: Working with local brokers will provide maximum information about the Bay Ridge market. ● Lawful and Financial advisors: Providing advice on compliance with the laws of real estate and assisting in providing the best financial strategie ● Technology Vendors: Providing smooth processing by offering CRM sites, property listing websites, and marketing support.	Key Activities ● Target Bay Ridge by selective marketing to attract sellers and buyers to the area. ● Provide a good service that will retain clients and attract referrals. ● Periodically check on the market conditions, revising strategies as appropriate. Key Resources ● Talented employees, responsible for daily operations and relations with the clients. ● An office space that is easily accessible for clients. ● Technology products to automate marketing for the optimization of business processes.	Value Propositions ● Offer a more competitive commission rate at 4.50% to attract more clients while maintaining a high level of service. ● Focus on personalized service that meets the particular needs of every client for maximum satisfaction and loyalty.	Customer Relationships ● Offer personalized consultations and support for buying or selling. ● Stay connected with clients post-transaction to build long-term relationships. ● Encourage referrals from existing clients with incentives. Channels ● Utilize social media, Google Ads, and real estate listings for leads and brand attention. ● Build community relationships through sponsorships, events, and business partnerships.	Customer Segments ● Target homeowners looking to sell quickly and with minimal costs. ● Focus on families, professionals, investors in the Bay Ridge area, and ● Attract real estate investors seeking profitable opportunities in Brooklyn.
Cost Structure ● Maintain office space costs under \$15,000 per month, including rent and utilities. ● Keep employee costs within budget while gradually expanding the team as needed. ● Allocate budget to ongoing marketing efforts. ● Invest in technology to simplify operations and improve customer interactions.			Revenue Streams ● The main source of income is from buying and selling properties, with a commission of 4.00% to 6.00% per sale. ● providing property management services to investors for extra income. Additionally, ● Generate money from partnerships with mortgage brokers, lawful firms, and other services related to real estate transactions.	

Recommendations & Next Steps

The optimization model demonstrated that a real estate company can open an office space in Bay Ridge since they can generate a positive NPV profit of \$1.94 million NPV over eight quarters. However, to maintain long-term profitability, the company should take the following steps:

1. Expand Workforce Gradually:

- Hire one employee initially to increase market penetration by 0.5% to reach 6.75%. As the business grows, you can expand the workforce by hiring one additional employee.

2. Implement Technology Solutions:

- Implement a customer relationship management (CRM) system to track leads, sales, and client interactions.
- Use marketing automation tools to facilitate email campaigns and social media posts.

3. Monitor and Adjust the Business Model Quarterly:

- Due to the fluctuating sales forecasts over the eight quarters, review Bay Ridge market data quarterly to track changes in real estate prices, rent costs, and competitor strategies. Adjust commission rates and marketing tactics as necessary.

4. Explore Additional Revenue Streams:

- Build referral partnerships with mortgage brokers, attorneys, and other professionals to generate additional income and value for your clients.